SEAN BALBALE

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PROFILE

I got into engineering because I wanted to understand how things actually work — not just how to use them. That instinct runs through everything I build: bare-metal Assembly for an 8051 microcontroller, real-time C++ firmware on an Arduino platform, and a senior capstone built around a fatigue-prediction algorithm running entirely on a Nordic nRF52840 — no cloud offload, sub-100 ms latency. The constraint of the edge is something I find genuinely satisfying; it forces precision that higher-level abstractions let you avoid. On the other side of the stack, I owned a Spring Boot 2 to Spring Boot 3 framework migration across multiple production Java services at Bullhorn, and this summer I led a WebSocket-to-SQS migration for two production services on Sonos's Cloud Content team, tracing a silent delivery failure to an endpoint-routing defect along the way. The problems are different at each layer — but the methodology isn't.

Beyond implementation, I research the systems we are building toward. "Embodied Harms and Inferred Data: Redefining Privacy in Extended Reality" argues that existing legal frameworks are structurally inadequate for immersive environments that passively extract behavioral biometrics, and proposes a "Bodyright" framework that protects human autonomy by design rather than by policy. It won the Alumni Prize in English Composition at Trinity's 2026 Honors Day and is forthcoming in *The International Journal of Law, Ethics, and Technology*. Outside the lab, I compete on Trinity's varsity rowing team and serve as Technology Chair for the Accidentals. One is about power and synchronization at 5 a.m. on the Connecticut River; the other is about blending twelve voices into something that sounds like one. The underlying discipline is identical: precise timing, collective execution, and zero tolerance for drift.

EDUCATION

Trinity College — Hartford, CT *Expected May 2027*
Bachelor of Science in Electrical and Computer Engineering
Bachelor of Science in Computer Science
GPA: 3.70 / 4.00

Coursework Completed: Data Structures & Algorithms, Embedded Systems Design, Basic Economic Principles, Age of Augustus, Ancient Greek Philosophy, Microelectronic Circuits, Applied Linear Algebra, Calculus III, Intro to Computer Systems, Writing for a Digital World.

In Progress (Spring 2026): High-Performance Computing, Deep Learning, Software Engineering, International Trade, Measurements, Instrumentation & Analysis, Engineering Materials, Linear Circuit Theory (ENGR 212L).

Planned (Fall 2026): Automatic Control Systems, Mechanics I, Computer Science Seminar, Capstone Design I — CPSC 498 / ECE 483.

AP Credit Transferred: Computer Science A, English Composition & Literature, Physics C: Mechanics, Physics C: Electricity & Magnetism.

Extracurriculars: Varsity Athlete — Trinity College Men's Rowing (Division III); Singer & Technology Chair — The Trinity College Accidentals (premier a cappella ensemble).

Carnegie Mellon University — Pittsburgh, PA (Summer Program) *June – August 2022*
Completed university-level coursework: Fundamentals of Programming & Computer Science (15-112); Reasoning with Data.

Northeastern University — Boston, MA & London, UK *September 2023 – December 2024*
Attended Northeastern University for three semesters before transferring to Trinity College (Spring 2025). All coursework transferred.

Freshman Fall 2023 — Boston, MA: General Chemistry I, Cornerstone of Engineering 1, Calculus I for Science & Engineering, Musical Communities of Boston.

Freshman Spring 2024 — London, UK (Study Abroad): Fundamentals of Computer Science II (with Lab), Cornerstone of Engineering 2, Calculus II for Science & Engineering, British Drama/London Stage.

Sophomore Fall 2024 — Boston, MA: Discrete Structures (with Seminar), Circuits/Signals: Biomedical Applications, Embedded Design: Enabling Robotics, Differential Equations & Linear Algebra for Engineering, First-Year Seminar, Chorus.

Weston High School — Weston, MA *September 2019 – June 2023*
High School Diploma **GPA:** 3.85 / 4.00

Activities: Varsity Captain — Wayland-Weston Men's Rowing; Varsity Ski Team; Concert Choir; Jazz Choir.

TECHNICAL SKILLS

Languages	Java, Python, C, C++, TypeScript, JavaScript (Node.js), Assembly (8051, MIPS), Verilog, SQL, MATLAB, LaTeX
Backend	Spring Boot 3, Jakarta EE, JUnit 5, Mockito, Node.js, FastAPI, REST APIs
Frontend / Full-Stack	React, Next.js, Svelte, Tailwind CSS, AngularJS
Cloud & DevOps	AWS, Kubernetes, Docker, Proxmox, Linux, Git (feature branches, merge requests, code review), CI/CD pipelines, Log4j2, Supabase, Slurm, MPI
Embedded / Hardware	Arduino, ESP32, 8051 Microcontroller, Intel Cyclone V FPGA, Nordic nRF52840, SPI/I^2C, BLE 5, ANT+, FreeRTOS, NI-DAQ, Oscilloscopes, Logic Analyzers, LTspice
Mechanical / CAD	SolidWorks, FDM 3D Printing (Klipper firmware), Laser Cutters, Precision Metrology Calibration
Other	Systems Engineering, Zero-Knowledge Architectures, Object-Oriented Design, Parallel Computing, MCP (Model Context Protocol) Server Development

WORK EXPERIENCE

Cloud/Service Software Engineering Intern — *Sonos* Summer 2026
Boston, MA

- Led migration of two production Java/Spring Boot services on the Cloud Content team from WebSocket command delivery to an SQS-backed fire-and-forget queue: authored the design plan and Jira epic, implemented feature-flagged per-player fan-out with retry-until-acceptance semantics, and validated both transports against real speakers.
- Diagnosed a silent delivery failure in a pre-production environment by tracing commands through the SDK and queue infrastructure to an endpoint-routing defect, and drove the fix across both services.
- Made the service's functional-test suite runnable locally by porting its DynamoDB-stream Lambda into a local test harness, enabling pre-push validation ahead of the shared CI run.
- Built Datadog dashboards and monitors for the new delivery path to support on-call visibility after launch.

Engineering Teaching Assistant — ENGR 212L — *Trinity College* Spring 2026
Hartford, CT

- Instructed students through weekly laboratory sessions for Linear Circuit Theory (ENGR 212L), covering node and mesh analysis, natural and forced responses of first- and second-order RLC circuits, operational amplifiers, and passive and active filter design.
- Guided students through the application of Laplace transforms and Fourier series as analytical tools for solving linear circuit equations, reinforcing lecture content with hands-on hardware experimentation.
- Delivered objective data analysis feedback on student lab write-ups to reinforce core ECE measurement and instrumentation practices.

Software Engineering Intern — *Bullhorn* Summer 2025
Boston, MA

- Contributed to enterprise-scale backend Java services across the Shield and Builders teams, supporting the SEARCH_AI, ATS_CRM, and MIDDLE_OFFICE engineering groups.
- Owned a major framework migration (BH Boot 1.0.0): upgraded multiple production Java applications from Spring Boot 2 to Spring Boot 3, systematically refactoring all `javax` namespaces to `jakarta` and restructuring the proprietary data access layer.
- Led a critical security initiative to upgrade core logging infrastructure to Log4j2, hardening the dependency chain across affected services.
- Refactored legacy test suites from TestNG/JMockit to JUnit 5 and Mockito, increasing test coverage and long-term maintainability.
- Participated in the full software development lifecycle: managed feature branches, authored merge requests, and conducted peer code reviews.

Metrology Intern — *East Coast Metrology — A Division of In-Place Machining Company* Summer 2024
Topsfield, MA

- Conducted in-house and on-site calibrations for high-performance precision measurement equipment, including portable CMMs, total stations, large-volume laser scanners, and laser trackers.
- Measured manufactured engineering parts against client SolidWorks CAD models to verify dimensional compliance with specified tolerances.
- Updated and maintained internal-use SolidWorks CAD models to reflect as-calibrated equipment configurations.

Engineering Intern — *Titan Advanced Energy Solutions* Summer 2023
Salem, MA

- Designed and implemented Python software to parse raw JSON output from ultrasound sensors and render waveform visualizations, providing a complementary view to existing heat-map outputs for non-destructive evaluation of electric vehicle battery packs.
- Visualization tooling directly accelerated the detection and classification of physical defects within battery cells, supporting quality assurance workflows.

Apprentice Optician — *Warby Parker* June 2023 – January 2024
Boston, MA

- Validated prescriptions, took optical measurements, and sold eyewear in a high-volume retail environment; worked part-time (5 hours per week) during the academic year and full-time during summer 2023.
- Ranked highest in sales for the Northeast region for three of the eight weeks worked full-time during the summer.
- Resolved customer complaints and maintained a high-quality client experience under pressure; developed skills in sales, customer service, and teamwork.

Founder & IT Consultant — *Cherrybrook Networks* March 2019 – May 2025
Weston, MA

- Founded and operated a freelance software development and technical support business serving local companies, students, and families in the greater Weston, MA area.
- Installed and configured software applications; designed and assembled custom computers tailored to client specifications.
- Deployed residential and small-business Wi-Fi infrastructure; automated secure cloud backup systems for client data protection.
- Maintained long-term client relationships over six years, developing skills in client communication, project scoping, and independent problem-solving.

Engineering Teaching Assistant — *Weston Public Schools* October 2022 – May 2023
Weston, MA

- Developed and delivered course curriculum for high school engineering and physics courses; wrote and graded assessments and provided individualized academic support.
- Maintained and calibrated precision workshop equipment, including FDM 3D printers, laser cutters, band saws, and woodworking tools.

Rowing Coach — *Wayland-Weston Rowing Association* September 2019 – June 2023
Wayland, MA

- Coached middle school athletes in on-water rowing technique during spring and fall crew seasons, building a foundation in catch timing, blade work, and race-pace conditioning.
- Built team culture and improved athletes' physical and mental fitness across four competitive seasons.

RESEARCH & PUBLICATIONS

"Embodied Harms and Inferred Data: Redefining Privacy in Extended Reality"
The International Journal of Law, Ethics, and Technology *In Production — Expected Late May 2026*

- Authored a peer-reviewed synthesis paper examining the political economy of Extended Reality (XR) technology and its implications for privacy law and human autonomy.
- Argued that existing legislative frameworks (BIPA, GDPR) are structurally inadequate for immersive environments that passively collect behavioral biometrics — including physiological signals, gaze patterns, and locomotion data — without active user consent.

- Deconstructed the "inference gap" — the mechanism by which immersive architectures bypass cognitive defenses to extract sensitive psychometric traits (health status, political orientation, sexual orientation) without user awareness.
- Conducted a comparative analysis of the Apple–Meta dichotomy, demonstrating that market-led "privacy-by-design" functions primarily as an anti-competitive moat, producing a "Walled Garden" paradox in which privacy becomes a luxury good.
- Applied the legal doctrines of "undue influence" and "defect of consent" to the "Proteus Effect" to establish that XR environments possess a unique capacity to vitiate user autonomy through subliminal behavioral modification.
- Proposed a transition from the "notice-and-choice" legal paradigm to a "Bodyright" framework, reclassifying biometric data as an inalienable extension of the human sovereign rather than an extractable raw material.

HONORS & AWARDS

Alumni Prize in English Composition — *Trinity College — Department of English* May 2026

- Received the Alumni Prize in English Composition at the Trinity College Honors Day Ceremony (May 1, 2026), awarded by the Department of English faculty to recognize the best expository writing produced in the year.
- Prize awarded for "Embodied Harms and Inferred Data: Redefining Privacy in Extended Reality," accepted for publication in *The International Journal of Law, Ethics, and Technology* (Expected Late May 2026).

SELECTED PROJECTS

Edge-Compute Pacing & Feedback Node (Senior Capstone) *Fall 2026 – Spring 2027 CPSC 498 / ECE 483 — with Zoe Davidow, advised by Prof. Kenneth Kousen*

- An embedded edge-compute node that delivers real-time biomechanical and fatigue feedback to endurance athletes (rowers and cyclists) without cloud or smartphone offload.
- Hardware platform: Nordic nRF52840 (ARM Cortex-M4F @ 64 MHz, 256 kB RAM, 1 MB Flash) with native dual-mode BLE 5 and ANT+ radio; WS2812B NeoPixel LED strip and piezo buzzer as output peripherals; 18650 Li-Ion cell with BMS for $\geq$6 h runtime.
- Core algorithm: discrete-time W' balance model (Skiba et al., 2012) implemented as a first-order Euler update in C/C++ on FreeRTOS, targeting $\leq$100 ms end-to-end sensor-to-LED latency on a bare-metal microcontroller.
- Ingests live power and heart-rate data via ANT+ Bicycle Power Profile (PWR) and Heart Rate Profile (HRM); BLE GATT power service as fallback.
- Open research questions: (1) algorithmic correctness under Euler discretisation vs. continuous ground truth; (2) optimal FreeRTOS scheduling policy for worst-case latency on a single-core MCU; (3) feasibility of a lightweight LSTM for personalised $\tau_{W'}$ estimation beyond population-level constants.

Astraphe AI — Mobile AI Athletic Coaching Platform *2026 – Active*

- Built a mobile-first AI coaching platform end-to-end: a multi-source data pipeline (Strava, WHOOP, Garmin, intervals.icu) deduplicating activity data onto canonical records, and a server-side training-load engine (TSS, CTL/ATL/TSB, composite Readiness, Strain) implemented in FastAPI.
- Built a Gemma 4 coaching agent with a pgvector RAG pipeline over coaching memory, exposed through a Svelte 5 / Capacitor mobile client (iOS/Android).
- Self-hosted the platform on a Kubernetes (k3s) cluster with CI/CD via GitHub Actions enforcing an 80% test-coverage gate.
- Later added a standalone MCP server exposing user data to Claude via a spec-compliant OAuth 2.1 flow (Dynamic Client Registration, PKCE), and open-sourced the platform under Apache-2.0.

Personal Portfolio Website *Spring 2026*

- Architected and deployed a production full-stack web application at seanbalbale.com using Next.js, React, and TypeScript; styled with Tailwind CSS and modern UI component libraries.
- Built a custom backend API route to handle contact form submissions and programmatically dispatch transactional emails; implemented dark/light mode theming and an embedded PDF resume viewer with dynamic project routing.
- Deployed via Vercel with CI/CD on each Git push.

Remote MCP Server (Obsidian Vault Integration) *Spring 2026*

- Built a TypeScript MCP (Model Context Protocol) server that exposes a personal Obsidian knowledge vault over a secure remote endpoint.

- Implemented Bearer token authentication and deployed via a Cloudflare Tunnel / Tailscale network overlay; orchestrated with Docker Compose multi-profile deployment supporting multiple environment configurations.

HPC Cluster — High-Performance Computing Lab *Spring 2026*

- Assembled and configured a personal three-node HPC cluster using Lenovo ThinkStation P340 SSF machines interconnected via a dedicated Gigabit Ethernet switch.
- Installed and configured Slurm workload manager for job scheduling; benchmarked parallel workloads using MPI for CPSC 375 (High-Performance Computing).

Cherrybrook Homelab — Self-Hosted Infrastructure *2025 – Present*

- Built and operate a four-node Proxmox VE cluster (custom i7-9700K + Tesla V100 server, two Dell PowerEdge servers, a repurposed ThinkPad quorum/backup node) hosting Dockerized production workloads, including Astraphe AI and a self-hosted Second Brain MCP server.
- Executed a live VLAN/subnet migration with split-horizon DNS to resolve VPN conflicts; manages reverse proxy/SSL, tailnet DNS, and a UniFi network stack.

Project Aletheia — Interactive XR Privacy Manifesto *Fall 2025*

- Built a full-stack React web application serving as an interactive privacy manifesto, demonstrating concrete biometric tracking risks inherent in Extended Reality environments.
- Advocates for Privacy-by-Design principles and client-side Zero-Knowledge architectures as a technical implementation of the "Bodyright" framework developed in published research.

Embedded Heart Rate Monitor (8051 Assembly) *Spring 2025*

- Designed and wired the complete hardware system for a clinical heart rate monitor and mechanical click counter on a custom breadboard.
- Wrote low-level Assembly language firmware for an 8051 microcontroller handling external analog input reading, interrupt-driven signal processing, ADC sampling, and real-time BCD display output — entirely without a C runtime or operating system.

Haptic Horizon — Wearable Navigation Device for the Visually Impaired *Spring 2024*

- Designed and built a wearable haptic alarm system to assist visually impaired users in navigating physical environments.
- Wrote embedded C++ firmware for an Arduino-based platform integrating an HC-SR04 ultrasonic ranging sensor and a haptic motor; sensor output encoded distance and direction into distinct vibration patterns without any auditory output.
- Handled interrupt-driven sensor polling, signal conditioning, and real-time actuator control within a single-threaded embedded event loop.

Robotic Control via FPGA *Fall 2024*

- Implemented a Verilog hardware description on an Intel Cyclone V FPGA to control a robotic system, demonstrating direct hardware-software co-design principles.
- Designed logic modules for motor control signals, sensor interfacing, and real-time feedback within the timing and resource constraints of the target FPGA fabric.

Content-Aware Java Image Editor — Seam Carving (CS2510 Term Project) *Spring 2024*

- Implemented the seam carving algorithm (Avidan & Shamir, 2007) for content-aware image resizing in Java, dynamically identifying and removing low-energy pixel seams to resize images while preserving salient visual content.
- Built an interactive command-line menu supporting multiple editing operations; applied object-oriented design patterns throughout using Java and Apache Maven.

"Fail Early and Often" — Python Roguelite (CMU 15-112 Term Project) *Summer 2022*

- Designed and implemented a fully-featured roguelite RPG in Python as the term project for CMU CS 15-112 (Fundamentals of Programming & CS); approximately 25 hours of development.
- Built core game systems: procedural map generation, WASD movement, proximity-based melee combat, a multi-slot inventory system with consumable potions, enemy AI wave logic, sprite rendering, collision detection, and a toggle-able cone-of-vision mechanic.
- Demo: youtu.be/aftX408l35E

Space Invaders — Python Arcade Game *2020*

- Designed and implemented a Space Invaders clone in Python using Pygame, practicing sprite rendering, collision detection, enemy AI wave logic, and a high-score persistence system.

ADDITIONAL PROJECTS & LINKS

Selected additional repositories from github.com/sbalbale (42 repositories, 91 stars) not detailed above:

- **Bike Handlebar Clamp** (reverse-engineering pipeline) — Reverse-engineered a replacement handlebar clamp via an iPhone photogrammetry → RealityCapture → CloudCompare → SolidWorks pipeline, designing a swept-solid geometry with a TPU-liner interface.
- **Trinity Accidentals Digital Heritage Platform** (trinity-accidentals) — Next.js / Sanity monorepo archiving the Trinity College Accidentals' performance history and media.
- **hermes-sendblue-plugin** (GitHub) — Sendblue (iMessage/SMS) integration plugin for the Hermes personal-agent framework.
- **boop-agent** (GitHub) — iMessage personal agent built on the Claude Agent SDK, with memory, sub-agents, and automations.
- **wealthfolio-simplefin-sync** (GitHub) — Addon syncing investment holdings and cash balances into Wealthfolio via the SimpleFIN Bridge.
- **3D-Printer-Firmware** (GitHub) — Custom Klipper firmware configuration for a modified Creality Ender 3 v2.

Repository links for projects detailed above: AstrapheAI (astraphe-ai-coach, renamed from "Astrape AI" May 2026), Cherrybrook Homelab (cherrybrook-networks), Remote MCP Server (second-brain-mcp), Personal Portfolio (portfolio-site), Project Aletheia (project-aletheia), Haptic Horizon (Haptic-Horizon).

ACTIVITIES & INTERESTS

Athletics: Varsity Rowing, Trinity College Men's Crew (Division III). Outside of crew, training interests include triathlon, road and gravel cycling (power meter training, pedal-stroke symmetry analysis), ski mountaineering, technical mountaineering, and backcountry backpacking.

Cooking: Serious home cook across a wide range of cuisines — tacos, samosas, katsu ramen, Indian, Italian, and whatever fusion experiment sounds right that week. The kitchen is where precision and improvisation meet, which is probably why it keeps me coming back.

Music & Technology: Technology Chair and Singer, Trinity College Accidentals (premier a cappella group) — manage live sound engineering, recording infrastructure, and digital production for performances and studio sessions.

Engineering & Making: 3D printing on a heavily modified Creality Ender 3 v2 running Klipper open-source firmware.

Research Interests: Privacy law (GDPR, BIPA, biometric autonomy); XR ethics and surveillance capitalism; hardware-constrained software design; distributed systems and cloud microservices; HPC and parallel computing; embedded systems for sports science.

REFERENCES

Available upon request.